Neural Networks 206 (2027) 109792

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

Bayesian nonparametric survival analysis via deep Dirichlet process

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ARTICLE INFO

Keywords:
Bayesian nonparametric methods
Big data
Deep learning
Machine learning
Survival analysis
Variational inference

ABSTRACT

Time-to-event data analysis has attracted growing interest across various domains. A major challenge lies in the inherent incompleteness of such data, with right censoring being the most common form. Although Cox's proportional hazards model is effective for traditional datasets, it often fails in modern settings where high-dimensional covariates violate the proportional hazards assumption. Additionally, conventional parametric survival models, typically based on the exponential family, impose strong assumptions and struggle to capture long-tail behavior due to their exponential decay. To address the aforementioned challenges, we propose DDPSurv, a novel deep learning framework for survival analysis that leverages a deeply parameterized Dirichlet Process (DP) mixture model. Unlike prior deep parametric approaches that rely on strong statistical assumptions, DDPSurv offers greater flexibility in modeling survival distributions by incorporating a DP mixture, enabling improved representation of heavy-tailed behavior. We provide theoretical guarantees showing that the model can closely approximate the true survival distribution with a tight concentration rate. Empirical results on standard survival benchmarks demonstrate the effectiveness of our method, while large-scale experiments on MIMIC-III and MIMIC-IV datasets further validate its scalability and clinical utility. Codes are available at https://github.com/HKU-MedAI/DDPSurv-net.

1. Introduction

One of the key challenges that differentiate time-to-event data from other types of data is the incomplete data issue and the censoring mechanism, including right-censoring (Cox, 1972; Nagpal et al., 2021b), and interval censoring (Zhang & Yin, 2023). Survival analysis, a branch of time-to-event modeling, deals with right-censored data in most cases and has found applications in various domains, such as clinical trials (Cox, 1972; Zhang & Yin, 2023) and actuarial science (Logubayom & Yeboah, 2023). Despite the success of traditional survival analysis statistical models (Gu et al., 2023; Li et al., 2018), they are incapable of capturing the significant features when scaling to modern high-dimensional datasets. Therefore, it is emerging to introduce deep learning approaches to time-to-event modeling to deal with high-dimensional data (Faraggi & Simon, 1995; Katzman et al., 2018; Xiang et al., 2000), as deep neural networks allow for a more flexible nonlinear relation between covariates and hazard ratio. Although deep neural networks have significantly improved survival models' prediction accuracy and flexibility, they now face the new challenge of being over-parameterized, leading to potential over-fitting issues. Recently, probabilistic models have been incorporated into the learning objective to regularize the learning process, where the deep Cox model (Nagpal et al., 2021b) is a well-known method that integrates the Cox proportional hazards (CPH) model and deep learning. However, these existing models rely on strong parametric assumptions (e.g., exponential family) and lack flexibility in modeling the survival distribution. Especially when the observations in tails are limited and the exponential tail decay would mostly lead to bias at long tail. Moreover, these works are heavily based on the CPH assumption with a negative partial likelihood to address the challenge of missing data, making it difficult to handle the cross-hazard scenario that is very common in real data analysis (Mantel & Stablein, 1988). In this paper, we propose DDPSurv, a deep Dirichlet process model with heavy tail mixtures, to estimate the survival distribution more accurately and efficiently. Exploiting the richer-gets-richer property of the Dirichlet process, our framework autonomously determines the appropriate number of mixture components and identifies the most prominent distribution from an infinite set, thereby providing an optimal approximation of the survival distribution. To address the long-tail bias inherent in conventional exponential-family distributions, we incorporate heavy-tailed distributions into the final mixture model. This is accomplished by introducing an additional Dirichlet process, which determines the optimal number of heavy-tailed distributions to be included. Our contributions can be summarized as follows: 1. We propose a new neural survival analysis framework (DDPSurv) based on a deep Dirichlet

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https://doi.org/10.1016/j.neunet.2026.109792
Received 21 July 2025; Received in revised form 16 May 2026; Accepted 18 October 2026
Available online 23 October 2026
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process, which can tackle survival prediction at high dimensions. 2. By mixing heavy tail distributions, we validate that the DP model can better approximate the survival distribution at tails and mitigate the long-tail bias. DDPSurv can also tackle the competing risk scenarios since each distribution in the infinite mixture can model the hazard rate of a particular risk. 3. DDPSurv adopts stochastic variational inference to avoid the high computational cost in parameter estimation via existing sampling-based methods (Müller et al., 2015; Zhang & Yin, 2023) (e.g., Gibbs sampling), and enables scalability to large-scale datasets. 4. Theoretical analysis under the Sieve space framework provides asymptotic bounds to the posterior concentration rate of the parameters. 5. Extensive experiments on generic survival predictions and two large-scale clinical datasets validate the satisfactory performance of our proposed method.

2. Related works

2.1. Survival analysis

Time-to-event modeling, particularly in the presence of censoring data, has been an important topic in statistical prediction across various domains such as economics (Bosco Sabuhoro et al., 2006; Jones et al., 2002), actuarial science (Czado & Rudolph, 2002), and medical treatment (Kim et al., 2019; Zhu et al., 2016). Survival analysis, a major subfield of time-to-event modeling, has been extensively studied. Two major traditional parametric or semi-parametric survival models that have played a prominent role in survival analysis are the Cox proportional hazard model (CPH) (Cox, 1972) and the accelerated failure time model (AFT) (Wei, 1992). A substantial body of literature (Kraisangka & Druzdzel, 2016, 2018; Rosen & Tanner, 1999) has focused on improving these models to achieve higher prediction performance.

In recent years, deep neural networks and stochastic variational inference methods have been applied to enhance traditional parametric or semi-parametric survival analysis (Alaa & van der Schaar, 2017; Katzman et al., 2018; Nagpal et al., 2021b; Zhong et al., 2021) to further improve estimation performance. While the deep neural networks framework increases the flexibility of the model and improves its capacity to handle high-dimensional data problems, stochastic variational inference allows the model to backpropagate gradients and thus save computational costs. DeepHit (Lee et al., 2018) and Deep survival machines (DSM) (Nagpal et al., 2021a) have successfully learned fully parametric models while employing full loss likelihood and stochastic variational inference respectively. However, these models still have limitations due to their fixed parameter settings or model assumptions including the number of mixture components in DSM and discrete-time cases and single-death causes in DeepHit. DSM, in particular, is known for its ability to handle competing risks by learning shared representations. Noting the limitations of previous parametric deep learning models for survival analysis, neural frailty machine (NFM) (Wu et al., 2023) manages to build a fully parameterized deep learning model based on frailty-based statistic models and provides robust statistical theoretical analysis for its convergence of prediction bias. NFM does not use a mixture model structure and still lacks flexibility when approximating survival functions.

2.2. Non-parametric analysis

Non-parametric models have played a crucial role in statistical analysis, offering flexibility and wide applicability (Peterson, 2009; Satagopan et al., 2004; Steinwart & Christmann, 2008). In the field of survival analysis, traditional non-parametric methods are mainly frequentist methods, including the Kaplan–Meier (KM) estimator (Kaplan & Meier, 1958) and the Nelson–Aalen estimator (Nelson, 1969). The KM estimator approximates the survival function by adjusting for the observed event times in its immediate neighborhood. However, frequentist methods ignore the prior knowledge and have a limited function search space compared with Bayesian methods. While Bayesian methods alleviate the limitations of parametric modeling and allow for a larger search space of functions. Among these Bayesian non-parametric methods, the Dirichlet process combined with the Gibbs sampling method has recently been used to solve the survival problems for its outstanding performance in clustering (Zhang & Yin, 2023) due to the richer-gets-richer property. However, this method still relies on a traditional statistical approach using a Dirichlet process of small-size parameters, without incorporating deep neural networks, which may limit its ability to handle complex features and high-dimensional data. Additionally, the Gibbs sampler may perform worse than the variational inference method in terms of computing efficiency. Therefore, stochastic variational inference needs to be adopted to incorporate non-parametric model into modern deep learning settings.

2.3. Long-tail bias correction

The problem of long-tail bias has been challenging for many real datasets in insurance, healthcare, and survival analysis (Fackrell, 2009; Hakim et al., 2021). Since the observations at tails are rather limited in most of the scenarios, it is difficult to approximate the distributions at tails. Previous works have employed a large number of parametric distributions belonging to an exponential family (Gardiner et al., 2014; Hakim et al., 2021) to handle the survival distribution when the data are heavy-tailed. However, common primitive distributions used in survival analysis (e.g., Weibull, log-normal) have poor tail behaviors due to their exponential tail decay (Landsman & Tsanakas, 2012).

Recently, two trends tackling the drawbacks of exponential family distributions have been proposed. One is to re-balance the dataset before the model learns representations by oversampling the tail data, augmenting tail data, or under-sampling the head data (Beery et al., 2020; Buda et al., 2018). Another one is to solve the long-tail bias by re-weighting the loss, setting the loss to be non-uniform, to facilitate learning the tail data (Cui et al., 2019; Samuel & Chechik, 2021). These approaches mainly focus on adjusting the dataset and the loss. They only provide a universal correction on the distribution and are still dominated by the exponential tail decay (e.g., Nagpal et al., 2021a). Therefore, a more flexible correction that can adaptively determine the density adjustment is needed for a more accurate approximation at tails.

3. Methodology

3.1. Preliminaries

3.1.1. Problem: time-to-event modeling

We assume a right-censoring model for simplicity. Let $\mathcal{D} = \{(\boldsymbol{x}_i, t_i, \delta_i)\}_{i=1}^n$ denote the dataset as a set of tuples, where $\boldsymbol{x}_i \in \mathbb{R}^d$ is the features associated with individual i, t_i is the time at which an event of interest occurs, or the time of censorship, and δ_i is the indicator that specifies whether t_i is the event time or censoring time. We denote the uncensored subset of $\mathcal{D}$ as $\mathcal{D}_U$ and the censored subset as $\mathcal{D}_C$.

Definition 3.1 (Dirichlet Process)**.** Denoted as $\mathrm{DP}(\alpha, G)$, the Dirichlet process is a random probability measure on the sample space $\mathcal{X}$, such that for any measurable finite partition of S, denoted as $\{B_i\}_{i=1}^K$,

$$(X(B_1), X(B_2), \ldots, X(B_K)) \sim \mathrm{Dir}(\alpha G(B_1), \alpha G(B_2) \ldots, \alpha G(B_K))$$

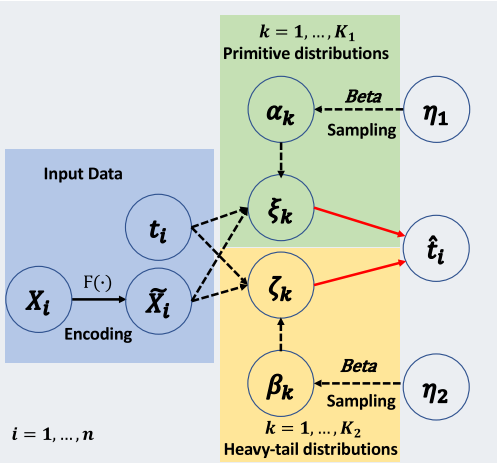


Fig. 1. Our proposed survival model in plate notations.

Definition 3.2 (Dirichlet Process Mixture (DPM))**.** Let DP(αG_0) denote a Dirichlet process with parameter αG_0 where α is a precision parameter and G_0 is a base probability distribution. T is the truncated number of mixtures. The DPM model is defined as

$$G \sim \mathrm{DP}(\alpha G_0),$$
$$\theta_1, \ldots, \theta_T \sim G,$$
$$x_k|\theta_k \sim f_{\theta_k},$$

Definition 3.3 (Heavy-tail Distribution)**.** We consider a distribution $F(x)$ as heavy-tail if its tail convergence rate is slower than an exponential decay, as given by

$$\int e^{tx} dF(x) = \infty, \text{ for all } t > 0.$$

3.2. Deep Dirichlet process mixture models

3.2.1. Overview

Fig. 2 demonstrates an overview of our proposed model DDPSurv, which provides an approximation for the survival curve for individuals by utilizing the Dirichlet Process to mix primitive distributions and heavy-tailed distributions. The features ($\tilde{X}$), encoded by covariates (X) via embedding networks ($F(\cdot)$), together with survival time t and censor indicator δ, are input into the K_1 primitive distributions and K_2 heavy-tail distributions. The weights for mixing model (π_k) is derived by stick-breaking process from sampled weights (α_k) via the Beta distribution. The derived weights π_k is the possibility assigned to each cluster and we denote Z as the cluster assignment indicator. The entire model is trained and optimized using variational inference and the evidence lower bound (ELBO). Details of loss settings and formulations can be found in Section 3.3. Meanwhile, the detailed procedures for training sets of networks can be found in Algorithm 1, and the plate notation in Fig. 1 provides a supplementary illustration of the relationships between the variables. The dashed lines in Fig. 1 indicate the model's parameters updating and the red lines represent the prediction section.

3.2.2. Dirichlet process

We represent the survival function with a mixture of primitive distributions (e.g., Log-Normal, Weibull), where detailed description of common primitive distributions is provided in the Appendix B.1 We specify the Dirichlet process mixture model with the well-known stick-breaking formulation,

$$\alpha_k \sim \mathrm{Beta}(1, \eta_1), \qquad \pi_k = \alpha_k \prod_{l=1}^{k-1}(1 - \alpha_l), \tag{1}$$

Algorithm 1 Our proposed DDPSurv framework.

Input:
Data $\mathcal{D} = \{(\widetilde{x}_i, t_i, \delta_i)\}_{i=1}^n$
K_1, the maximum number of primitive distributions;
K_2, the maximum number of heavy-tail distributions;
Parameter sets of primitive distributions ξ and heavy-tail distributions ζ.
Hyperparameters $\{\eta_1, \eta_2\}$,
Parameter sets of networks $\Psi = \{\psi_k\}_{k=1}^{K_1}$ and $\Upsilon = \{\upsilon_k\}_{k=1}^{K_2}$ to encode the shape and scale parameters of the primitive distributions and heavy-tail distributions, respectively.
Output: Trained Ψ and Υ

Sample π_k with Eq. (1)
for each training epoch **do**
  for $g \in \{1, \ldots, K_1\}$ **do** ▷ Primitive Distributions
    Sample mixture weights by Eq. (1) with concentration rate η_1
    Encode parameters ξ with Ψ
    Compute log-likelihood with Eq. (2)
  end for
  for $g \in \{1, \ldots, K_2\}$ **do**
    Sample mixture weights by Eq. (4) with concentration rate η_2
    Encode parameters ζ with Υ
    Compute log-likelihood with Eq. (3)
  end for
  Compute ELBO with Eq. (8)
  Backpropagate ELBO to Ψ and Υ
end for
return Trained Ψ and Υ.

where π_k is the probability assigned to each cluster. We further assume for each individual we either observe the actual failure time or censoring time but not both. Then we have $z|\boldsymbol{\pi} \sim \mathrm{Cat}(\cdot|\boldsymbol{\pi})$ the cluster assignment sampled from the multinomial distribution based on the cluster probability $\boldsymbol{\pi}$, and η_1 is the concentration hyperparameter. We truncate the number of primitive distributions at $K = K_1$.

3.2.3. Primitive distributions

Let $\xi = \{\xi_k\}_{k=1}^{K_1}$ represent the set of all shape and scale parameters of the primitive distribution. We adopt the log-normal distribution as an illustrative example for the primitive distribution where $\xi_k = (\vartheta_k, \varsigma_k)$. Then density $f(t|\xi)$ and the survival function $S(t|\xi)$ are given by

$$f(t|\xi_k) = \frac{1}{t\varsigma_k\sqrt{2\pi}} e^{-\frac{(\log t - \vartheta_k)^2}{2\varsigma_k^2}}$$
$$S(t|\xi_k) = 1 - \frac{1}{2}\mathrm{erfc}\left(-\frac{\log t - \vartheta_k}{\sqrt{2}\varsigma_k}\right) \tag{2}$$

Further discussions on the primitive distributions are presented in Appendix B.1.

3.2.4. Mitigating long-tail bias

As the existing distributions based on the exponential family suffer from exponential tail decay and poorly model the tail behavior of the true survival distribution, we further include K_2 heavy-tail distributions into the mixture to improve the tail behavior (Dey & Yan, 2016; Ibragimov et al., 2015). Without loss of generality, we adopt the log-Cauchy distribution with density function

$$f(x; \mu, \sigma) = \frac{1}{x\pi\sigma\left[1 + \left(\frac{\log x - \mu}{\sigma}\right)^2\right]}, \tag{3}$$

which has a logarithmically decaying tail. In addition to the mixture of primitive distributions, we mix infinite numbers of heavy tail distributions as specified by Eq. (3). We then apply another stick-breaking

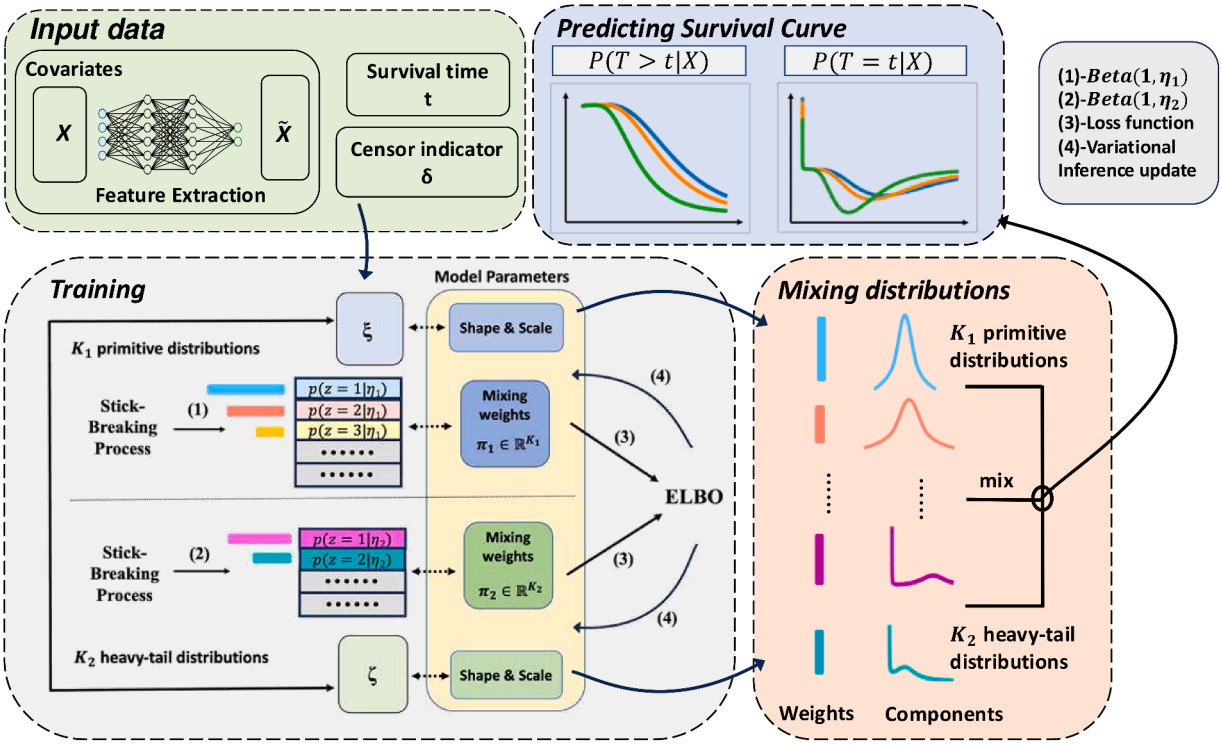


Fig. 2. The workflow of our proposed survival analysis framework. We first encode the scale and shape parameters of the primitive distributions. We then adopt the stick-breaking formulation of the Dirichlet process to determine the mixture weights of each mixture component. The mixed distribution is then employed to approximate the conditional survival curve of the individual.

process to compute the mixture weights $\{\lambda_k\}_{k=1}^{K_2}$,

$$\beta_k \sim \text{Beta}(1, \eta_2), \qquad \lambda_k = \beta_k \prod_{l=1}^{k-1}(1-\beta_l). \tag{4}$$

Let $\boldsymbol{\zeta} = \{\boldsymbol{\zeta}_k\}_{k=1}^{K_2}$ represent the sets of all shape and scale parameters of the heavy-tail distributions, where $\boldsymbol{\zeta}_k$ is the set of parameters of the kth mixture. We further denote $\boldsymbol{\phi} = \{\boldsymbol{\xi}, \boldsymbol{\zeta}\}$.

3.3. Stochastic variational inference

We adopt stochastic variational inference to train our proposed model and provide estimates to $\boldsymbol{\phi}$. We derive respectively the log likelihood (as the negation of ELBO) for the uncensored part and censored part. Without loss of generality, we focus on the right-censoring scheme, and we specify the censoring and uncensoring loss functions as follows.

3.3.1. Uncensoring loss

Given the DP mixture model, we have the following loss for uncensored data,

$$\begin{aligned}\log \mathbb{P}(D_U|\Theta) &= \log\left(\prod_{i=1}^{|D_U|} P(T=t_i|\boldsymbol{x}_i,\Theta)\right)\\ &= \sum_{i=1}^{|D_U|}\log\left(\sum_{k=1}^{K_1} P(T=t_i|\boldsymbol{x}_i,\boldsymbol{\xi}_k)P(Z=k|\boldsymbol{x}_i,\beta)\right)\\ &+ \sum_{i=1}^{|D_U|}\log\left(\sum_{k=K_1+1}^{K_1+K_2} P(T=t_i|\boldsymbol{x}_i,\boldsymbol{\zeta}_k)P(Z=k|\boldsymbol{x}_i,\beta)\right)\end{aligned} \tag{5}$$

3.3.2. Censoring loss

For right-censored data, the ELBO is defined by the likelihood to the survival function

$$\begin{aligned}\log \mathbb{P}(D_C|\Theta) &= \log\left(\prod_{i=1}^{|D_C|} P(T>t_i|\boldsymbol{x}_i,\Theta)\right)\\ &= \sum_{i=1}^{|D_C|}\log\left(\sum_{k=1}^{K_1} P(T>t_i|\boldsymbol{x}_i,\boldsymbol{\xi}_k)P(Z=k|\boldsymbol{x}_i,\beta)\right)\\ &+ \sum_{i=1}^{|D_C|}\log\left(\sum_{k=K_1+1}^{K_1+K_2} P(T>t_i|\boldsymbol{x}_i,\boldsymbol{\zeta}_k)P(Z=k|\boldsymbol{x}_i,\beta)\right)\end{aligned} \tag{6}$$

By adopting a mean-field approximation, the variational family of the parameters is given by

$$\begin{aligned}q(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\xi},\boldsymbol{\zeta},t) = &\prod_{k=1}^{K_1-1} q(\alpha_k)\prod_{k=1}^{K_2-1} q(\beta_k)\prod_{k=1}^{K_1} q(\boldsymbol{\xi}_k)\\ &\cdot\prod_{k=1}^{K_2} q(\boldsymbol{\zeta}_k)\prod_{i=1}^{n} q(t_i|\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\xi},\boldsymbol{\zeta}),\end{aligned} \tag{7}$$

where q is the variational posterior, p is the assumed prior distribution.

The surrogate loss of backpropagation can be obtained by negating the evidence lower bound, which is computed by the log of posterior mixture weights,

$$\begin{aligned}\mathcal{L}(\boldsymbol{\Psi}) = &\ \text{KL}(q(\boldsymbol{\xi})\|p(\boldsymbol{\xi})) + \text{KL}(q(\boldsymbol{\zeta})\|p(\boldsymbol{\zeta}))\\ &+ \text{KL}(q(\boldsymbol{\alpha})\|p(\boldsymbol{\alpha})) + \text{KL}(q(\boldsymbol{\beta})\|p(\boldsymbol{\beta}))\\ &+ \sum_i \text{KL}(q(t_i|\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\xi},\boldsymbol{\zeta})\|p(t_i|\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\xi},\boldsymbol{\zeta})).\end{aligned} \tag{8}$$

Here, $\text{KL}(q(t_i|\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\xi},\boldsymbol{\zeta})\|p(t_i|\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\xi},\boldsymbol{\zeta}))$ is the censored likelihood which is specified by Eqs. (5) and (6), then the problem is reduced to learning the likelihood of inputs to the assumed DP model. The detailed parameter update procedure of the variational posterior and the closed-form KL divergences of the primitive distributions can be found in Appendix B.3.

4. Theoretical analysis

For technical simplicity, we assume that the hazard rate after mixture is predicted by the neural network, i.e.,

$$h(t|\boldsymbol{x}) = \nu(t,\boldsymbol{x}).$$

Table 1
Summary of datasets.

Dataset	Type	No. Obs.	Feature Dim.	No. Events	No. Censoring
SUPPORT	Single Risk	9105	38	6,201(68.11%)	2,904(31.89%)
SYNTHETIC	Multiple Risk	5000	9	4,003(80.06%)	997(19.94%)
MIMIC-III	Single Risk	17,814	34	2,235(12.55%)	14598(87.45%)
MIMIC-IV	Single Risk	22,913	30	2,703(11.80%)	20210(88.20%)

Here ν is an unspecified non-negative function. The survival function can then be represented by $S(t|X) = e^{-\int_0^t e^{\nu(s,\boldsymbol{x})}ds}$. Then the censored log-likelihood can be re-written as

$$l(T,\delta,\boldsymbol{x};\nu) = \delta \log \int_0^T e^{\nu(s,\boldsymbol{x})}ds + \delta\nu(t,\boldsymbol{x}) + \int_0^T e^{\nu(s,\boldsymbol{x})}ds.$$

We demonstrate the theoretical boundedness of the proposed DDPSurv model using the Sieve space (Wellner et al., 2013; Wu et al., 2023), which provides the rates of convergence in the sense of parametric regression. As in previous works (Wellner et al., 2013; Wu et al., 2023), we choose the Hölder ball to represent the function space,

$$W_M^\beta(\mathcal{X}) = \left\{ F : \max_{\alpha:|\alpha|\le\beta} \operatorname*{ess\,sup}_{x\in\mathcal{X}} |D^\alpha(f(x)) \le M \right\},$$

where the domain $\mathcal{X}$ is assumed to be a subset of d-dimensional euclidean space, $\alpha = (\alpha_1,\dots,\alpha_d)$ is a d-dimensional tuple of nonnegative integers satisfying $|\alpha| = \alpha_1 + \cdots + \alpha_d$ and $D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1}\cdots x_d^{\alpha_d}}$ is the weak derivative of f. We assume M is a reasonably large constant.

We make the following assumption for the true parameters:

Condition 1 (True Parameter Wu et al., 2023)**.** The Euclidean parameter $\theta_0 \in \Theta \subset \mathbb{R}$, and the two function parameters $m_0 \in W_M^\beta(\mathcal{X})([-1,1]^d)$, $h_0 \in W_M^\beta(\mathcal{X})([0,\tau])$, and $\tau > 0$ is the ending time in the theoretical studies in survival analysis.

Condition 2 (Sieve space)**.** The Sieve space $\mathcal{V}_n$ is constructed as a set of MLPs satisfying $\hat{\nu} \in \mathcal{W}_{M_v}^\beta([0,\tau])$, with depth of order $\mathcal{O}(\log n)$ and total number of parameters $\mathcal{O}\left(n^{\frac{d+1}{\beta+d+1}}\log n\right)$. Here, M_v is a sufficiently large constant such that every function in $\mathcal{W}_{M_v}^\beta([0,\tau])$ can be accurately approximated by functions inside $\mathcal{V}_n$.

Let ν_0 be the true parameter and $\hat{\nu}$ be the corresponding estimate. We define $\mathbb{P}_{\hat{\nu}_n,\boldsymbol{x}}$ to be the estimated conditional distribution given $\boldsymbol{x}$ and $\mathbb{P}_{\nu_0,\boldsymbol{x}}$ to be the true conditional distribution. We further define a metric to measure the convergence of the parameter estimate,

$$d(\hat{\nu}_n,\nu_0) = \sqrt{\mathbb{E}_{x\sim\mathbb{P}_X}\left[H^2\left(\mathbb{P}_{\hat{\nu}_n,\boldsymbol{x}}\|\mathbb{P}_{\nu_0,\boldsymbol{x}}\right)\right]}, \tag{9}$$

where $H^2\left(\mathbb{P}_{\hat{\nu}_n,\boldsymbol{x}}\|\mathbb{P}_{\nu_0,\boldsymbol{x}}\right) = \int(\sqrt{d\mathbb{P}} - \sqrt{d\mathbb{Q}})^2$ is the squared Hellinger distance between the probability distributions $\mathbb{P}$ and $\mathbb{Q}$. We use $\tilde{\mathcal{O}}$ to hide the poly-logarithmic factors in the big-O notation.

Based on the above regularity conditions of the Sieve space, we can state the following theorem on the rate of convergence.

Theorem 1 (Rate of convergence)**.** *Under Conditions 1 and 2, we have that* $d(\hat{\nu}_n,\nu_0) = \tilde{\mathcal{O}}\left(n^{-\frac{\beta}{2\beta+2d+2}}\right)$.

More detailed proofs can be viewed in Appendix A.

5. Experiments

5.1. Datasets

We validate our method on two common datasets for survival prediction — SUPPORT and SYNTHETIC. We additionally include two large-scale benchmarks on real-world clinical data — the MIMIC-III dataset and the MIMIC-IV dataset. The datasets size and basic information including: Risk type, number of observation, feature dimension, number of events and number of censoring are given in Table 1. Other key characteristics of these datasets are summarized in Appendix C.1.

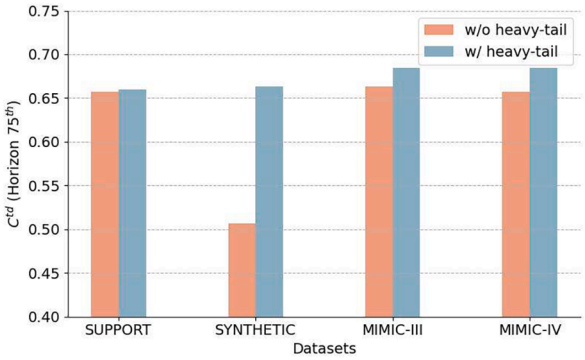


Fig. 3. The performance in C^{td} of DDPSurv under 75th time horizon with and without mixing the heavy-tail distributions, respectively.

5.2. Evaluation metric

We use two standard metrics in survival analysis for evaluating model performance. One is the **time-dependent Concordance-Index (C^{td})** (Antolini et al., 2005):

$$C^{td} = \mathbb{P}(\hat{F}(t|x_i) > \hat{F}(t|x_j)|\delta_i = 1, T_i < T_j, T_i \le t),$$

where $\hat{F}(t|X)$ is the estimated CDF by the model at the truncation time t, given features X. An unbiased probability estimate is obtained by performing pairwise comparisons of relative risks and applying inverse-probability-of-censoring weighting, following standard practice in survival analysis (Gerds et al., 2013). The other metric is **Brier score (BS)**, $\text{BS} = \frac{1}{N}\sum_{t=1}^N (f_t - o_t)^2$ where f_t is the predicted probability of the event, o_t is the actual outcome of the event at instance t and N is the number of forecasting instances. Smaller BS value indicates good performance. Detailed descriptions of the reason that we use time dependent C^{td} instead of Harrell's C-index as evaluation metrics are provided in the Discussion section.

5.3. Compared methods

We compare our proposed framework to six competitors: **(1) Cox Proportional Hazards (CPH)** (Cox, 1972): This is the standard semi-parametric model, making the assumption of constant baseline hazard. The features interact with the learnt set of weights in a log-linear fashion in order to determine the hazard for a held-out individual. **(2) DeepHit (DH)** (Lee et al., 2018): This approach involves learning the joint distribution of all event times by jointly modeling all competing risks and discretizing the output space of event times. **(3) Deep Cox Mixture (DCM)** (**Nagpal** et al., **2021b**)**:** This model replaces the parameters in CPH with deep neural networks and adopts a mixture model structure. **(4) Deep Survival Machines (DSM)** (Nagpal et al., 2021a): This model is mixture of parametric models from lognormal or Weibull distributions. It does not rely on the strong assumption of Cox proportional hazard ratio. **(5) Sumo-Net** (Rindt et al., 2022): This model proposes a simple novel survival regression method using a monotonic restriction on the time-dependent weights to optimize right-censored log-likelihood. **(6) NFM** (Wu et al., 2023): This model proposes a fully parameterized deep learning model based on the frailty model.

5.4. Implementation details and hyperparameters

The proposed method is implemented in Python with *Pytorch* library on a server equipped with four NVIDIA GeForce RTX 3090 GPUs. All

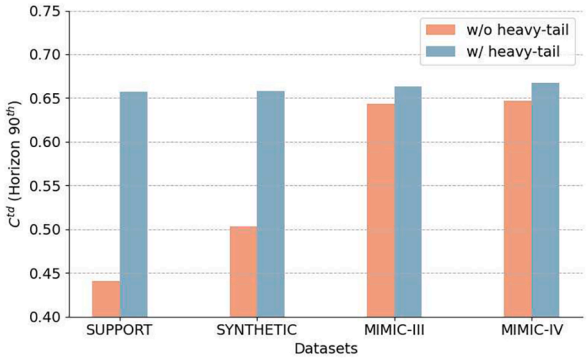


Fig. 4. The performance in C^{td} of DDPSurv under 90th time horizon with and without mixing the heavy-tail distributions, respectively.

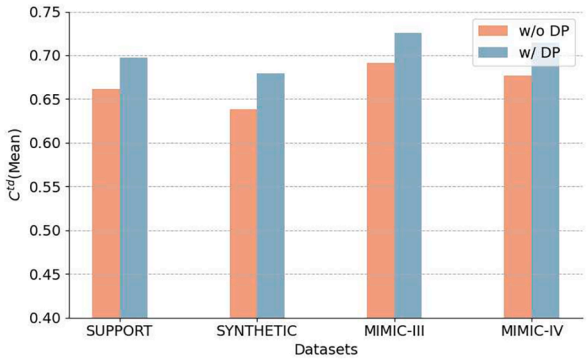


Fig. 5. Ablation on the effectiveness of Dirichlet process. (with and without DP setting).

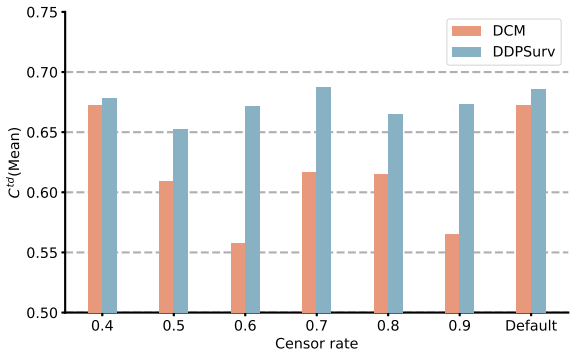


Fig. 6. The mean of four time horizons' C^{td} under different censor rates.

models are pre-trained with 10,000 iterations and then trained with 100 epochs. We use the *Adam* optimizer to optimize the model with a learning rate of 1×10^{-4}. The representation learning function $F(\cdot)$ is a fully connected multi-layer perceptron (MLP) with one or two hidden layers and ReLU6 activations. Based on the empirical results, we set the default network size to 500 for SUPPORT and SYNTHETIC dataset and set the default network size to 2000 to MIMIC-III and MIMIC-IV dataset. The concentration rate is set to 10. The detailed ablation results on these hyperparameters can be viewed in Appendix C.2.

5.5. Quantitative results

We evaluate our model and baseline methods at four different time horizons: the 25th, 50th, 75th, and 90th quantiles. Performance at the first two time horizons reflects the model's general survival prediction ability, while the latter two focus on its tail performance.

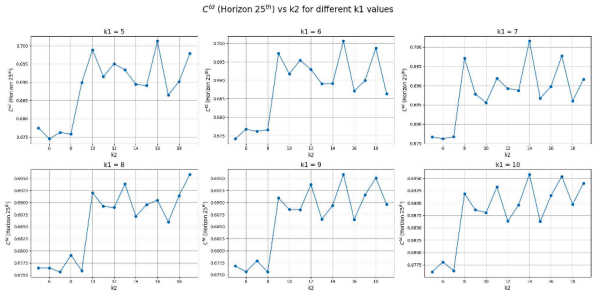


Fig. 7. Performance(C^{td} for 25th time horizon) of our DDPSurv on the MIMIC-III dataset with respect to different K_1 and K_2.

5.5.1. Performance on 25th and 50th time horizons

The results are shown in Table 2. Our model outperforms the baseline methods with a notable margin in most settings on both C^{td} and the Brier score metrics. DDPSurv ranks first in 14 out of 16 cases and second in the remaining two. Notably, the model demonstrates more significant improvements on the MIMIC-III and MIMIC-IV datasets, with increases of approximately 1.03% and 2.41%, respectively, compared to the SUPPORT (0.28%) dataset. This underscores the model's effectiveness in addressing high censoring rates, which present challenges for many prior approaches. Furthermore, the performance gains on large-scale datasets such as MIMIC-III and MIMIC-IV emphasize the scalability of the DDPSurv model.

5.5.2. Performance on 75th and 90th time horizons

For evaluating tail performance after incorporating the heavy-tail mixture, we assess the model on the 75th and 90th quantiles. Table 3 shows the performance of DDPSurv at these time horizon quantiles. Our model ranks first in 12/16 cases and second or third in the remaining instances. The performance gain in C^{td} for the tail quantiles (1.58% performance gain for C^{td} across all datasets) is larger than that for the 25th and 50th quantiles (0.75% performance gain for C^{td} across all datasets), indicating that DDPSurv effectively handles tail scenarios and mitigates long-tail bias. Moreover, DDPSurv outperforms other mixture-model baselines (such as DSM and DCM), suggesting that the heavy-tail mixture enhances tail performance. We further validate the impact of the heavy-tail mixture by comparing DDPSurv's performance with and without the mixture and Figs. 3 and 4 display the results for the 75th and 90th time horizons, respectively. The version without the heavy-tail mixture is implemented by setting the hyperparameter K_2 to 0. We observe a 15% improvement in prediction performance for the SYNTHETIC dataset and approximately 2% for MIMIC-III and MIMIC-IV datasets with the heavy-tail mixture, using C^{td} as the evaluation metric. This trend is consistent across all datasets, with the SUPPORT dataset at the 90th time horizon showing the most substantial improvement, exceeding 20%.

5.6. Ablation analysis

5.6.1. Effectiveness of Dirichlet process

To show the effectiveness of the Dirichlet process on the survival prediction, we simply compare the performance of without DP setting variant version and our complete DDPSurv to estimate the effects of using the Dirichlet Process on mixture models. To exclude the effect of mixing heavy-tail distributions, we set $K_2 = 0$ and only mix primitive distributions. We examine the mean C^{td} of four time horizons under different datasets to show the effects. As shown in Fig. 5, DDPSurv outperforms the variant without DP by a considerable margin of around 3–4% on all four datasets, indicating that using the Dirichlet Process on mixture models improves the prediction performance.

5.6.2. Effects of the censoring rate

We further investigate the effect of the censoring rate on model performance by conducting ablation experiments on the SUPPORT dataset,

Table 2
Compared results at 25th time horizon and 50th time horizon. **Best results** across the comparable methods in each dataset are highlighted in bold, while the second-best results are underlined.

25th Time Horizon	SUPPORT		SYNTHETIC		MIMIC-III		MIMIC-IV	
Models	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓
CPH (Cox, 1972)	$68.52_{\pm 0.00}$	$48.54_{\pm 0.00}$	$62.66_{\pm 0.00}$	$36.76_{\pm 0.00}$	$76.41_{\pm 0.00}$	$4.87_{\pm 0.00}$	$71.97_{\pm 0.00}$	$4.67_{\pm 0.00}$
DeepHit (Lee et al., 2018)	$62.90_{\pm 0.40}$	$18.50_{\pm 1.09}$	$61.84_{\pm 0.76}$	$18.78_{\pm 7.77}$	$71.49_{\pm 0.64}$	$28.63_{\pm 1.56}$	$70.48_{\pm 0.64}$	$46.37_{\pm 1.60}$
DCM (Nagpal et al., 2021b)	$\underline{76.40}_{\pm 0.99}$	$11.58_{\pm 0.31}$	$67.35_{\pm 0.30}$	$15.89_{\pm 0.21}$	$80.50_{\pm 1.16}$	$4.05_{\pm 0.05}$	$\underline{75.70}_{\pm 1.03}$	$4.19_{\pm 0.04}$
DSM (Nagpal et al., 2021a)	$75.90_{\pm 0.41}$	$\underline{11.17}_{\pm 0.04}$	$67.69_{\pm 0.28}$	$15.99_{\pm 0.03}$	$\underline{81.84}_{\pm 0.51}$	$\underline{3.93}_{\pm 0.02}$	$75.18_{\pm 1.26}$	$\underline{4.16}_{\pm 0.02}$
Sumo-Net (Rindt et al., 2022)	$64.64_{\pm 2.08}$	$28.87_{\pm 0.47}$	$65.43_{\pm 1.75}$	$30.90_{\pm 0.04}$	$64.09_{\pm 0.30}$	$22.21_{\pm 0.53}$	$64.59_{\pm 0.13}$	$54.85_{\pm 0.80}$
NFM (Wu et al., 2023)	$69.91_{\pm 4.01}$	$30.72_{\pm 0.20}$	$\underline{67.74}_{\pm 0.44}$	$\mathbf{15.34}_{\pm 0.12}$	$69.09_{\pm 0.09}$	$59.14_{\pm 0.08}$	$68.65_{\pm 0.02}$	$62.57_{\pm 0.50}$
DDPSurv	$\mathbf{76.82}_{\pm 0.34}$	$\mathbf{11.13}_{\pm 0.03}$	$\mathbf{70.16}_{\pm 0.38}$	$\underline{15.44}_{\pm 0.18}$	$\mathbf{82.73}_{\pm 0.84}$	$\mathbf{3.91}_{\pm 0.03}$	$\mathbf{79.15}_{\pm 0.48}$	$\mathbf{4.10}_{\pm 0.01}$
50th Time Horizon	SUPPORT		SYNTHETIC		MIMIC-III		MIMIC-IV	
Models	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓
CPH (Cox, 1972)	$66.50_{\pm 0.00}$	$34.34_{\pm 0.00}$	$60.73_{\pm 0.00}$	$23.47_{\pm 0.00}$	$69.63_{\pm 0.00}$	$10.23_{\pm 0.00}$	$71.34_{\pm 00.00}$	$11.41_{\pm 00.00}$
DeepHit (Lee et al., 2018)	$63.51_{\pm 0.76}$	$24.43_{\pm 0.47}$	$68.09_{\pm 0.38}$	$33.04_{\pm 9.05}$	$71.07_{\pm 0.54}$	$29.43_{\pm 0.74}$	$70.28_{\pm 0.54}$	$38.16_{\pm 0.74}$
DCM (Nagpal et al., 2021b)	$\underline{70.76}_{\pm 0.60}$	$19.04_{\pm 0.54}$	$67.23_{\pm 0.11}$	$23.62_{\pm 0.42}$	$71.36_{\pm 0.99}$	$9.95_{\pm 0.12}$	$68.57_{\pm 0.47}$	$\underline{10.58}_{\pm 0.07}$
DSM (Nagpal et al., 2021a)	$70.19_{\pm 0.34}$	$\underline{18.33}_{\pm 0.07}$	$66.69_{\pm 0.28}$	$24.10_{\pm 0.08}$	$\underline{72.98}_{\pm 0.70}$	$\underline{9.66}_{\pm 0.06}$	$\underline{72.86}_{\pm 1.24}$	$10.59_{\pm 0.05}$
Sumo-Net (Rindt et al., 2022)	$64.64_{\pm 2.08}$	$29.94_{\pm 0.47}$	$64.64_{\pm 2.09}$	$32.19_{\pm 0.82}$	$66.21_{\pm 0.35}$	$14.26_{\pm 0.07}$	$64.56_{\pm 0.22}$	$36.90_{\pm 4.13}$
NFM (Wu et al., 2023)	$63.18_{\pm 0.18}$	$40.49_{\pm 0.15}$	$\underline{69.30}_{\pm 0.21}$	$\mathbf{20.32}_{\pm 0.12}$	$68.67_{\pm 0.07}$	$51.14_{\pm 0.02}$	$67.30_{\pm 0.11}$	$51.11_{\pm 0.03}$
DDPSurv	$\mathbf{70.89}_{\pm 0.64}$	$\mathbf{18.17}_{\pm 0.03}$	$\mathbf{69.71}_{\pm 0.42}$	$\underline{20.93}_{\pm 0.35}$	$\mathbf{74.15}_{\pm 0.19}$	$\mathbf{9.45}_{\pm 0.03}$	$\mathbf{74.23}_{\pm 0.24}$	$\mathbf{10.31}_{\pm 0.07}$

Table 3
Compared results at the tail, i.e., 75th time horizon and 90th time horizon. **Best results** across each dataset are in bold, while the second-best results are underlined.

75th Time Horizon	SUPPORT		SYNTHETIC		MIMIC-III		MIMIC-IV	
CPH (Cox, 1972)	$\underline{66.02}_{\pm 0.00}$	$23.15_{\pm 0.00}$	$59.74_{\pm 0.00}$	$49.34_{\pm 0.00}$	$65.02_{\pm 0.00}$	$22.77_{\pm 0.00}$	$\underline{67.72}_{\pm 0.00}$	$27.27_{\pm 0.00}$
DeepHit (Lee et al., 2018)	$64.26_{\pm 0.73}$	$23.98_{\pm 0.28}$	$57.75_{\pm 5.68}$	$28.60_{\pm 03.34}$	$61.71_{\pm 0.44}$	$21.59_{\pm 2.07}$	$64.97_{\pm 0.44}$	$25.12_{\pm 3.00}$
DCM (Nagpal et al., 2021b)	$65.67_{\pm 1.89}$	$22.48_{\pm 0.54}$	$67.41_{\pm 0.08}$	$19.90_{\pm 0.17}$	$\underline{66.94}_{\pm 0.86}$	$18.14_{\pm 0.29}$	$66.90_{\pm 0.70}$	$19.85_{\pm 0.19}$
DSM (Nagpal et al., 2021a)	$65.47_{\pm 0.27}$	$\underline{22.02}_{\pm 0.10}$	$66.07_{\pm 0.32}$	$17.08_{\pm 0.49}$	$66.51_{\pm 0.38}$	$\underline{17.37}_{\pm 0.04}$	$67.65_{\pm 1.52}$	$\underline{19.69}_{\pm 0.19}$
Sumo-Net (Rindt et al., 2022)	$64.64_{\pm 2.08}$	$27.09_{\pm 0.90}$	$63.49_{\pm 2.09}$	$26.37_{\pm 3.12}$	$55.97_{\pm 9.02}$	$39.00_{\pm 10.94}$	$60.69_{\pm 4.86}$	$25.25_{\pm 3.11}$
NFM (Wu et al., 2023)	$63.67_{\pm 0.07}$	$34.06_{\pm 0.14}$	$\underline{68.54}_{\pm 0.11}$	$\mathbf{15.50}_{\pm 0.06}$	$66.28_{\pm 0.23}$	$31.90_{\pm 0.03}$	$66.48_{\pm 0.26}$	$32.36_{\pm 0.02}$
DDPSurv	$\mathbf{67.68}_{\pm 0.01}$	$\mathbf{21.86}_{\pm 0.08}$	$\mathbf{69.51}_{\pm 0.73}$	$\underline{15.84}_{\pm 0.29}$	$\mathbf{69.78}_{\pm 0.36}$	$\mathbf{16.72}_{\pm 0.05}$	$\mathbf{69.28}_{\pm 0.36}$	$\mathbf{19.30}_{\pm 0.10}$
90th Time Horizon	SUPPORT		SYNTHETIC		MIMIC-III		MIMIC-IV	
Models	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓	C^{td}(%) ↑	BS(%) ↓
CPH (Cox, 1972)	$\underline{65.92}_{\pm 0.00}$	$19.41_{\pm 0.00}$	$59.52_{\pm 0.00}$	$74.41_{\pm 0.00}$	$63.94_{\pm 0.00}$	$35.61_{\pm 0.00}$	$65.52_{\pm 0.00}$	$46.72_{\pm 0.00}$
DeepHit (Lee et al., 2018)	$64.19_{\pm 0.51}$	$24.56_{\pm 0.38}$	$57.35_{\pm 7.75}$	$14.43_{\pm 0.58}$	$64.57_{\pm 1.35}$	$\mathbf{13.64}_{\pm 0.47}$	$64.28_{\pm 0.05}$	$\underline{14.44}_{\pm 0.65}$
DCM (Nagpal et al., 2021b)	$65.16_{\pm 1.46}$	$17.81_{\pm 0.42}$	$\underline{67.02}_{\pm 0.16}$	$10.95_{\pm 0.22}$	$65.45_{\pm 0.56}$	$22.17_{\pm 0.20}$	$64.51_{\pm 1.09}$	$23.62_{\pm 0.40}$
DSM (Nagpal et al., 2021a)	$65.10_{\pm 0.20}$	$\underline{17.36}_{\pm 0.02}$	$65.08_{\pm 0.27}$	$10.19_{\pm 0.13}$	$\underline{65.89}_{\pm 0.41}$	$21.45_{\pm 0.14}$	$64.97_{\pm 1.89}$	$23.03_{\pm 0.55}$
Sumo-Net (Rindt et al., 2022)	$64.64_{\pm 2.08}$	$28.07_{\pm 0.86}$	$63.04_{\pm 3.66}$	$57.21_{\pm 3.08}$	$55.82_{\pm 8.76}$	$42.36_{\pm 9.96}$	$60.55_{\pm 4.86}$	$\mathbf{14.18}_{\pm 2.65}$
NFM (Wu et al., 2023)	$64.34_{\pm 0.08}$	$20.07_{\pm 0.05}$	$66.85_{\pm 0.11}$	$\mathbf{9.13}_{\pm 0.10}$	$64.79_{\pm 0.41}$	$\underline{16.29}_{\pm 0.00}$	$\underline{65.56}_{\pm 0.19}$	$16.83_{\pm 0.01}$
DDPSurv	$\mathbf{67.06}_{\pm 0.25}$	$\mathbf{17.28}_{\pm 0.05}$	$\mathbf{68.96}_{\pm 0.72}$	$\underline{9.41}_{\pm 0.13}$	$\mathbf{67.60}_{\pm \mathbf{0.28}}$	$20.54_{\pm 0.33}$	$\mathbf{66.38}_{\pm 0.20}$	$22.62_{\pm 0.13}$

which has a relatively low default censoring rate and allows for censoring rate adjustments. We adjust the dataset's censoring rate by keeping the censored data unchanged and randomly converting a portion of the uncensored data to censored. As shown in Fig. 6, we use the mean of C^{td} as the evaluation metric and compare the performance of our model with DCM, a strong baseline. The results show that our model is more robust to variations in the censoring rate, indicating a stable advantage over DCM across most censoring rates.

5.6.3. Number of mixture components

We evaluate the effect of different numbers of mixture components K_1 and K_2 on the survival prediction performance. As shown in Fig. 7, we use the C^{td} at 25th time horizon as the evaluation metric and run our experiments on the MIMIC-III dataset with different combinations of K_1 and K_2. The results indicate that the performance generally rises as a trend when K_2 increases within a range (from 5 to 10), which further suggests that a large number of heavy-tailed mixture components may improve the expressiveness of our model. While K_1 does not show significant impact on the model performance. The results stabilize after a certain number of heavy-tailed mixtures (e.g., $K_2 > 10$), indicating that the DP can automatically select the optimal number of mixture components, and hence reduce the reliance on tuning the number of mixtures.

6. Discussions

The survival datasets, including SUPPORT (Knaus et al., 1995), MIMIC-III (Johnson et al., 2016), and MIMIC-IV (Johnson et al., 2023), suffer from data imbalance, where the number of individuals with short survival times (head class) significantly exceeds those with long survival times (tail class). This imbalance can lead to overfitting, degrading the model's performance in predicting the tail of the survival function. By comparing the results in Table 2 (head class) and Table 3 (tail class), we observe that the C-index values for the first two time horizons are substantially higher than those for the 75th and 90th quantiles, suggesting that the scarcity of tail class data contributes to weaker tail predictions. Additionally, when comparing statistical methods (e.g., CPH) to deep learning-based methods, we find that deep learning approaches perform better for the more abundant data in the first two quantiles, while CPH outperforms many deep learning methods in tail scenarios. These findings imply that deep learning models require sufficient data

for effective training, and increasing the dataset size could improve the tail performance of our DDPSurv.

The Weibull distribution is commonly used in survival analysis due to its computational convenience and strong performance in practical applications. However, we opt to use the log-normal distribution, another widely used distribution, as our primary distribution. This choice is due to the gradient instability observed with the Weibull distribution during training, which negatively impacts performance across several datasets.

In our empirical experiments, we use the time-dependent C-index as the primary evaluation metric instead of Harrell's C-index. This choice allows us to assess how well the models capture changes in risk over time, thus addressing the assumption of constant proportional hazards made by the C-index. Meanwhile, this setting examines the model's performance on tail time horizons (75th and 90th time horizon) after incorporating heavy-tail distributions with the Dirichlet Process. Therefore, we present the time-dependent C-index at event horizon quantiles of 25th, 50th, 75th and 90th.

7. Conclusions

In this work, we propose DDPSurv, a novel deep Bayesian nonparametric framework on survival prediction. By mixing heavy-tail distributions into our model, we achieve adaptive tail correction and improve the behavior at tails. We adopt stochastic variational inference to train the model in high dimensions. Empirical results show that our method can overall outperform the baseline methods. Ablation analysis demonstrates the contribution of each proposed component and robustness to variations in hyperparameters. Our work can be potentially extended to multimodal learning, where Bayesian nonparametric methods can effectively fuse the distributions from different modalities.

CRediT authorship contribution statement

Jiacheng Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Formal analysis, Data curation; **Tsai Hor Chan:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization; **Fuying Wang:** Writing – review & editing, Software, Data curation; **Guosheng Yin:** Writing – review & editing, Supervision, Conceptualization; **Lequan Yu:** Writing – review & editing, Supervision, Resources, Project administration.

Limitations and future works

One limitation of our method is that we did not explicitly model the potential heterogeneity in individuals (although implicitly by adopting a mixture model). However, DDPSurv can be easily extended to incorporate this factor with modifications in the likelihood, such as the frailty family, which will be explored in future works. Our method can also be extended to other application domains, such as survival analysis for whole slide images.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT in order to polish the language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The codes for reproducing this work are available at https://github.com/HKU-MedAI/DDPSurv-net.

Appendix A. Technical details

Proof of Theorem 1. We follow the FN scheme from Wu et al. (2023) to prove the convergence properties of the proposed DDPSurv model. □

A.1. Technical lemmas

We list the technical lemma of the FN scheme here, where the proof of each technical lemma is extended from Wu et al. (2023). We first develop technical lemmas for facilitating the proof of the main theorem.

Lemma 1. *Under Conditions 1–2, for* $(T, \delta, \boldsymbol{x}) \in [0, \tau] \times \{0, 1\} \times [-1, 1]^d$ *the following terms are bounded:*

1. $l(T, \delta, \boldsymbol{x}; \nu_0)$ *with true parameter* ν_0.
2. $l(T, \delta, \boldsymbol{x}; \hat{\nu})$ *with any parameter estimates* $\hat{\nu}$ *in any Sieve space listed in Condition 2.*

Lemma 2. *Under Conditions 1–2, let* $\hat{\nu}$, $\hat{\nu}_1$, *and* $\hat{\nu}_2$ *be arbitrary three parameter tuples inside the sieve space defined in Condition 2, then the following inequalities hold*

$$\|l(T, \delta, \boldsymbol{x}; \nu_0) - l(T, \delta, \boldsymbol{x}; \hat{\nu})\|_\infty \lesssim \|\nu_0 - \hat{\nu}\|_\infty,$$
$$\|l(T, \delta, \boldsymbol{x}; \hat{\nu}_1) - l(T, \delta, \boldsymbol{x}; \hat{\nu}_2)\|_\infty \lesssim \|\hat{\nu}_1 - \hat{\nu}_2\|_\infty.$$

Lemma 3. *(Approximation error) For any* n, *there exists an element in the corresponding sieve space* $\varpi_n \nu_0$, *satisfying* $d(\varpi_n \nu_0, \nu_0) = \mathcal{O}(n^{-\frac{\beta}{\beta+d+1}})$.

Lemma 4. *Suppose that* $\mathcal{F}$ *is a class of functions satisfying that* $N(\epsilon, \mathcal{F}, \|\cdot\|) < \infty$ *for* $\forall \epsilon > 0$. *We define* $\tilde{N}(\epsilon, \mathcal{F}, \|\cdot\|)$ *to be the minimal number of* ϵ*-balls* $B(f, \epsilon) = \{g : \|g - f\| < \epsilon\}$ *needed to cover* $\mathcal{F}$ *and further constrain that* $f \in \mathcal{F}$. *Then we have*

$$N(\epsilon, \mathcal{F}, \|\cdot\|) \le \tilde{N}(\epsilon, \mathcal{F}, \|\cdot\|) \le N(\frac{\epsilon}{2}, \mathcal{F}, \|\cdot\|).$$

Lemma 5. *Suppose that* $\mathcal{F}$ *is a class of functions satisfying* $N(\epsilon, \mathcal{F}, \|\cdot\|_\infty) < \infty$ *for* $\forall \epsilon > 0$. *We define* $\tilde{N}_{[]}(\epsilon, \mathcal{F}, \|\cdot\|_\infty)$ *to be the minimal number of brackets* $[l, u]$ *needed to cover* $\mathcal{F}$ *with* $\|l - u\|_\infty \le \epsilon$ *and further constrain that* $f \in \mathcal{F}$, $l = f - \frac{\epsilon}{2}$, *and* $u = f + \frac{\epsilon}{2}$. *Then we have*

$$N_{[]}(\epsilon, \mathcal{F}, \|\cdot\|_\infty) \le \tilde{N}_{[]}(\epsilon, \mathcal{F}, \|\cdot\|_\infty) \le N_{[]}(\frac{\epsilon}{2}, \mathcal{F}, \|\cdot\|_\infty).$$

Lemma 6. *(Model Capacity) Let* $\mathcal{G}_n = \{l(T, \delta, \boldsymbol{x}; \hat{\nu}) : \hat{\nu} \in \mathcal{V}_n\}$. *Under Condition 2, with* $s_\nu = \frac{2\beta}{2\beta+d+1}$, *there exists a constant* $c_\nu > 0$ *such that*

$$N_{[]}(\epsilon, \mathcal{G}_n, \|\cdot\|_\infty) \lesssim \frac{1}{\epsilon} N(c_\nu \epsilon^{1/s_\nu}, \mathcal{V}_n, \|\cdot\|_2).$$

We adopt the theory of empirical processes (Wellner et al., 2013; Wu et al., 2023) heavily in the proof of main theorems. The proof is based on the proof of the FN scheme introduced by Wu et al. (2023). For a function class $\mathcal{F}$, we define $N(\epsilon, \mathcal{F}, \|\cdot\|)$ and $N_{[]}(\epsilon, \mathcal{F}, \|\cdot\|)$ to be the covering number and the bracketing number of $\mathcal{F}$ with respect to norm $\|\cdot\|$ under radius ϵ, respectively. We use $\mathrm{VC}(\mathcal{F})$ to denote the VC-dimension of $\mathcal{F}$. Moreover, we use the notation $a \lesssim b$ to denote $a \le Cb$ for some positive constant C.

We define

$$l(T, \delta, \boldsymbol{x}; \hat{\nu}) = \delta \log \int_0^T e^{\nu(s,\boldsymbol{x})} ds + \delta \nu(t, \boldsymbol{x}) + \int_0^T e^{\nu(s,\boldsymbol{x})} ds.$$

Under the definition of the sieve space stated in Condition 2, we restate the parameter estimate as

$$\hat{\nu}_n(t, \boldsymbol{x}) = \arg\max_{\hat{\nu} \in \mathcal{V}_n} \frac{1}{n} \sum_{i \in [n]} l(t_i, \delta_i, \boldsymbol{x}_i; \hat{\nu}).$$

Under the model assumption, $p(T,\delta|\boldsymbol{x}=\boldsymbol{x})$ can be expressed by

$$p(T,\delta|\boldsymbol{x};\nu)=\exp(l(T,\delta,\boldsymbol{x};\nu))f_{C|\boldsymbol{x}}(T)^{1-\delta}S_{C|\boldsymbol{x}}(T)^{\delta}.$$

The defined distance can be explicitly expressed by

$$d(\hat{\nu},\nu_0)=\sqrt{\mathbb{E}_{\boldsymbol{x}}\left[\int\left|\sqrt{p(T,\delta|\boldsymbol{x};\hat{\nu})}-\sqrt{p(T,\delta|\boldsymbol{x};\nu_0)}\right|^2\mu(dT\times d\delta)\right]}.$$

The proof can be then divided into four steps (Wu et al., 2023).

Step 1:

For arbitrary $0<\epsilon\leq 1$, we have that

$$\begin{aligned}&\inf_{d(\hat{\nu},\nu_0)\geq\epsilon}\mathbb{E}\left[l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu})\right]\\=&\inf_{d(\hat{\nu},\nu_0)\geq\epsilon}\mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{T|\delta,\boldsymbol{x}}\left[\log p(T,\delta|\boldsymbol{x};\nu_0)-\log p(T,\delta|\boldsymbol{x};\hat{\nu})\right]\right]\\=&\inf_{d(\hat{\nu},\nu_0)\geq\epsilon}\mathbb{E}_{\boldsymbol{x}}\left[\mathrm{KL}\left(\mathbb{P}_{\hat{\nu},\boldsymbol{x}}\|\mathbb{P}_{\nu_0,\boldsymbol{x}}\right)\right].\end{aligned}$$

Using the fact that $\mathrm{KL}\left(\mathbb{P}_{\hat{\nu},\boldsymbol{x}}\|\mathbb{P}_{\nu_0,\boldsymbol{x}}\right)\geq 2H^2\left(\mathbb{P}_{\hat{\nu},\boldsymbol{x}}\|\mathbb{P}_{\nu_0,\boldsymbol{x}}\right)$, we can further obtain that

$$\begin{aligned}&\inf_{d(\hat{\nu},\nu_0)\geq\epsilon}\mathbb{E}\left[l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu})\right]\\\geq&\inf_{d(\hat{\nu},\nu_0)\geq\epsilon}\mathbb{E}_{\boldsymbol{x}}\left[2H^2\left(\mathbb{P}_{\hat{\nu},\boldsymbol{x}}\|\mathbb{P}_{\nu_0,\boldsymbol{x}}\right)\right]\\=&2\inf_{d(\hat{\nu},\nu_0)\geq\epsilon}d^2(\hat{\nu},\nu_0)\\\geq&2\epsilon^2.\end{aligned}$$

Step 2: We consider the following derivations

$$\begin{aligned}&\sup_{d(\hat{\nu},\nu_0)\leq\epsilon}\mathrm{Var}\left[l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu})\right]\\\leq&\sup_{d(\hat{\nu},\nu_0)\leq\epsilon}\mathbb{E}\left[\left(l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu})\right)^2\right]\\=&\sup_{d(\hat{\nu},\nu_0)\leq\epsilon}\mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{T|\delta,\boldsymbol{x}}\left[\left(\log p(T,\delta,\boldsymbol{x};\nu_0)-\log p(T,\delta,\boldsymbol{x};\hat{\nu})\right)^2\right]\right]\\=&4\sup_{d(\hat{\nu},\nu_0)\leq\epsilon}\mathbb{E}_{\boldsymbol{x}}\left[\int\left(p(T,\delta,\boldsymbol{x};\nu_0)\left(\sqrt{\frac{p(T,\delta,\boldsymbol{x};\nu_0)}{p(T,\delta,\boldsymbol{x};\hat{\nu})}}\right)^2\right)\mu(dT\times d\delta)\right].\end{aligned}$$

By Taylor's expansion on $\log x$, there exists $\eta(T,\delta,\boldsymbol{x})$ between $\sqrt{p(T,\delta,\boldsymbol{x};\nu_0)}$ and $\sqrt{p(T,\delta,\boldsymbol{x};\hat{\nu})}$ pointwisely such that

$$\begin{aligned}&p(T,\delta,\boldsymbol{x};\nu_0)\left(\log\sqrt{\frac{p(T,\delta,\boldsymbol{x};\nu_0)}{p(T,\delta,\boldsymbol{x};\hat{\nu})}}\right)^2\\=&p(T,\delta,\boldsymbol{x};\nu_0)\left(\log\sqrt{p(T,\delta,\boldsymbol{x};\nu_0)}-\log\sqrt{p(T,\delta,\boldsymbol{x};\hat{\nu})}\right)^2.\\=&\frac{p(T,\delta,\boldsymbol{x};\nu_0)}{\eta(T,\delta,\boldsymbol{x})^2}\left(\sqrt{p(T,\delta,\boldsymbol{x};\nu_0)}-\sqrt{p(T,\delta,\boldsymbol{x};\hat{\nu})}\right)^2.\end{aligned}$$

Since $p(T,\delta,\boldsymbol{x};\nu_0)/p(T,\delta,\boldsymbol{x};\hat{\nu})=\exp(l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu}))$, from Lemma 1, $l(T,\delta,\boldsymbol{x};\nu_0)$ and $l(T,\delta,\boldsymbol{x};\hat{\nu})$ are bounded on $[0,\tau]\times\{0,1\}\times[-1,1]^d$ uniformly for all $\hat{\nu}$. Thus there exists constants C_1 and C_2 such that $0<C_1\leq p(T,\delta,\boldsymbol{x};\nu_0)/p(T,\delta,\boldsymbol{x};\hat{\nu})\leq C_2$. This leads to the fact that $p(T,\delta,\boldsymbol{x};\nu_0)\frac{1}{\eta(T,\delta,\boldsymbol{x})^2}$ is bounded. We further have that

$$\begin{aligned}&p(T,\delta,\boldsymbol{x};\nu_0)\left(\log\sqrt{p(T,\delta,\boldsymbol{x};\nu_0)}-\log\sqrt{p(T,\delta,\boldsymbol{x};\hat{\nu})}\right)^2\\\lesssim&\left|\sqrt{p(T,\delta,\boldsymbol{x};\nu_0)}-\sqrt{p(T,\delta,\boldsymbol{x};\hat{\nu})}\right|^2.\end{aligned}$$

Thus we have that

$$\begin{aligned}&\sup_{d(\hat{\nu},\nu_0)\leq\epsilon}\mathrm{Var}\left[l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu})\right]\\\lesssim&\sup_{d(\hat{\nu},\nu_0)}\mathbb{E}_{\boldsymbol{x}}\left[\int\left|\sqrt{p(T,\delta,\boldsymbol{x};\nu_0)}-\sqrt{p(T,\delta,\boldsymbol{x};\hat{\nu})}\right|^2\mu(dT\times d\delta)\right]\\=&\sup_{d(\hat{\nu},\nu_0)}d^2(\hat{\nu},\nu_0)\\\lesssim&\epsilon^2.\end{aligned}$$

Step 3 We define $\tilde{\mathcal{G}}_n=\{l(T,\delta,\boldsymbol{x};\hat{\nu})-l(T,\delta,\boldsymbol{x};\varpi_n\nu_0):\hat{\nu}\in\mathcal{V}_n\}$. Here $\varpi_n\nu_0$ has been defined in Lemma 3. Obviously, we have that $\log N_{[]}(\epsilon,\tilde{\mathcal{G}}_n,\|\cdot\|_\infty)=\log N_{[]}(\epsilon,\mathcal{G}_n,\|\cdot\|_\infty)$, where $\mathcal{G}$ is defined in Lemma 6. By Lemma 6, we further obtain that

$$N_{[]}(\epsilon,\mathcal{G}_n,\|\cdot\|_\infty)\lesssim\frac{1}{\epsilon}N(c_\nu\epsilon^{1/s_\nu},\mathcal{V}_n,\|\cdot\|_2).$$

According to Yarotsky (2017), Theorem 7, under Condition 2, we have that the VC-dimension of $\mathcal{V}_n$ satisfies that $\mathrm{VC}(\mathcal{V}_n)\lesssim n^{\frac{d+1}{\beta+d+1}}\log^3 n\log\frac{1}{\epsilon}$. Thus we obtain that

$$\log N(c_\nu\epsilon^{1/s_\nu},\mathcal{V}_n,\|\cdot\|)\lesssim\frac{\mathrm{VC}(\mathcal{V}_n)}{s_\nu}\log\frac{1}{\epsilon}\lesssim n^{\frac{d+1}{\beta+d+1}}\log^3 n\log\frac{1}{\epsilon}.$$

Furthermore, we also have that $\log N_{[]}(\epsilon,\tilde{\mathcal{G}},\|\cdot\|)\lesssim n^{\frac{d+1}{\beta+d+1}}\log^3 n\log\frac{1}{\epsilon}$.

Step 4 By the Cauchy–Schwarz inequality, we have that

$$\begin{aligned}&\sqrt{\mathbb{E}\left[l(T,\delta,\boldsymbol{x};\hat{\nu})-l(T,\delta,\boldsymbol{x};\varpi_n\nu_0)\right]}\\\leq&\left[\mathbb{E}(l(T,\delta,\boldsymbol{x};\hat{\nu})-l(T,\delta,\boldsymbol{x};\varpi_n\nu_0))^2\right]^{1/4}.\end{aligned}$$

Then by Lemma 3 we further obtain that

$$\begin{aligned}&\sqrt{\mathbb{E}\left[l(T,\delta,\boldsymbol{x};\hat{\nu})-l(T,\delta,\boldsymbol{x};\varpi_n\nu_0)\right]}\\\lesssim&\sqrt{d(\varpi_n\nu_0,\nu_0)}\lesssim n^{-\frac{\beta}{2\beta+2d+2}}.\end{aligned}$$

Now let

$$\tau=\frac{\beta}{2\beta+2d+2}-2\frac{\log\log n}{\log n}.$$

Then by steps 1, 2, 3 and Yarotsky (2017), Theorem 1,

$$d(\hat{\nu},\nu_0)=\max\left(n^{-\tau},d(\varpi_n\nu_0,\nu_0),\sqrt{\mathbb{E}\left[l(T,\delta,\boldsymbol{x};\hat{\nu})-l(T,\delta,\boldsymbol{x};\varpi_n\nu_0)\right]}\right).$$

By Lemma 3, we have $d(\varpi_n\nu_0,\nu_0)=\mathcal{O}(n^{-\frac{\beta}{\beta+d+1}})$, and by Step 4, we have $\sqrt{\mathbb{E}\left[l(T,\delta,\boldsymbol{x};\hat{\nu})-l(T,\delta,\boldsymbol{x};\varpi_n\nu_0)\right]}=\mathcal{O}(n^{-\frac{\beta}{2\beta+2d+2}})$. Thus we have $d(\hat{\nu},\nu_0)=\mathcal{O}(n^{-\frac{\beta}{2\beta+2d+2}}\log^2 n)=\tilde{\mathcal{O}}(n^{-\frac{\beta}{2\beta+2d+2}})$.

The proof of the technical lemmas is based on Wu et al. (2023).

Proof of Lemma 1. Since $\nu_0(T,\boldsymbol{x})\in\mathcal{W}_M^\beta([0,\tau]\times[-1,1]^d)$, we have that $\nu_0(T,\boldsymbol{x})\leq M$ and $\int_0^T e^{\nu(s,\boldsymbol{x})}ds\leq\tau e^M$.

$$\begin{aligned}&|l(T,\delta,\boldsymbol{x};\nu_0)|\\\leq&\left|\log\int_0^T e^{\nu_0(s,\boldsymbol{x})}ds\right|+|\nu_0(T,\boldsymbol{x})|+\left|\int_0^T e^{\nu_0(s,\boldsymbol{x})}ds\right|\\\leq&2M+\log\tau+\tau e^M.\end{aligned}$$

We then have that $l(T,\delta,\boldsymbol{x};\nu_0)$ is bounded among $(T,\delta,\boldsymbol{x})\in[0,\tau]\times\{0,1\}\times[-1,1]^d$. The proof of the boundedness of $l(T,\delta,\boldsymbol{x};\hat{\nu})$ is similar. □

Proof of Lemma 2. By definition we have that

$$\begin{aligned}&|l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu})|\\\leq&\left|\log\int_0^T e^{\nu_0(s,\boldsymbol{x})}ds-\log\int_0^T e^{\hat{\nu}(s,\boldsymbol{x})}ds\right|\\&+|\nu_0(T,\boldsymbol{x})-\hat{\nu}(T,\boldsymbol{x})|+\left|\int_0^T e^{\nu_0(s,\boldsymbol{x})}ds-\int_0^T e^{\hat{\nu}(s,\boldsymbol{x})}ds\right|.\end{aligned}$$

By Taylor's expansion on $\log(\cdot)$, we can further show that

$$\begin{aligned}&|l(T,\delta,\boldsymbol{x};\nu_0)-l(T,\delta,\boldsymbol{x};\hat{\nu})|\\\leq&|\nu_0(T,\boldsymbol{x})-\hat{\nu}(T,\boldsymbol{x})|+2\left|\int_0^T e^{\nu_0(s,\boldsymbol{x})}ds-\int_0^T e^{\hat{\nu}(s,\boldsymbol{x})}ds\right|.\end{aligned}$$

Again, by Taylor's expansion, we have

$$\left|\int_0^T e^{\nu_0(s,\boldsymbol{x})}ds-\int_0^T e^{\hat{\nu}(s,\boldsymbol{x})}ds\right|\leq\tau e^{\max(M,N_\nu)}\|\nu_0-\hat{\nu}\|_\infty.$$

Finally, we obtain that

$$|l(T,\delta,\boldsymbol{x};\nu_0) - l(T,\delta,\boldsymbol{x};\hat{\nu})| \le |\nu_0(T,\boldsymbol{x}) - \hat{\nu}(T,\boldsymbol{x})| + 2\tau e^{\max(M,N_\nu)}\|\nu_0 - \hat{\nu}\|_\infty.$$

Taking the supremum on both sides, we conclude that,

$$\|l(T,\delta,\boldsymbol{x};\nu_0) - l(T,\delta,\boldsymbol{x};\hat{\nu})\|_\infty \lesssim \|\nu_0 - \hat{\nu}\|_\infty.$$

The proof of the second inequality is similar. □

Proof of Lemma 3. According to Yarotsky (2017), Theorem 1, there exists an approximation function $\hat{\nu}^*$ such that $\|\nu_0 - \hat{\nu}^*\|_\infty = \mathcal{O}\left(n^{-\frac{\beta}{\beta+d+1}}\right)$. Let $\varpi_n\nu_0 = \hat{\nu}^*$. We have that

$$\begin{aligned}
&d(\varpi_n\nu_0, \nu_0)\\
&= \sqrt{\mathbb{E}_{\boldsymbol{x}}\left[\int \left|\sqrt{p(T,\delta|\boldsymbol{x};\hat{\nu})} - \sqrt{p(T,\delta|\boldsymbol{x};\nu_0)}\right|^2 \mu(dT\times d\delta)\right]}\\
&= \sqrt{\mathbb{E}_{\boldsymbol{x}}\left[\int \left[e^{\frac{1}{2}l(T,\delta,\boldsymbol{x};\varpi_n\nu_0)} - e^{\frac{1}{2}l(T,\delta,\boldsymbol{x};\nu_0)}\right]^2\right]}\\
&\cdot\sqrt{f_{C|\boldsymbol{x}}(T)^{1-\delta}S_{C|\boldsymbol{x}}(T)^{\delta}\mu(dT\times d\delta)}\\
&= \left\|e^{\frac{1}{2}l(T,\delta,\boldsymbol{x};\varpi_n\nu_0)} - e^{\frac{1}{2}l(T,\delta,\boldsymbol{x};\nu_0)}\right\|_\infty\\
&\cdot\sqrt{\mathbb{E}_{\boldsymbol{x}}\left[\int f_{C|\boldsymbol{x}}(T)^{1-\delta}S_{C|\boldsymbol{x}}(T)^{\delta}\right]}.
\end{aligned}$$

By Lemmas 1 and 2, we have that

$$\left\|e^{\frac{1}{2}l(T,\delta,\boldsymbol{x};\varpi_n\nu_0)} - e^{\frac{1}{2}l(T,\delta,\boldsymbol{x};\nu_0)}\right\|_\infty \le \|\varpi_n\nu_0 - \nu_0\|_\infty = \mathcal{O}\left(n^{-\frac{\beta}{\beta+d+1}}\right).$$

Since $f_{C|\boldsymbol{x}}(T)^{1-\delta} \le f_{C|\boldsymbol{x}}(T)$ and $S_{C|\boldsymbol{x}}(T)^{\delta} \le 1$, we also have that

$$\begin{aligned}
&\sqrt{\mathbb{E}_{\boldsymbol{x}}\left[\int f_{C|\boldsymbol{x}}(T)^{1-\delta}S_{C|\boldsymbol{x}}(T)^{\delta}\mu(dT\times d\delta)\right]}\\
&\le \sqrt{\mathbb{E}\left[(1+f_{C|\boldsymbol{x}}(T))\mu(dT\times d\delta)\right]}\\
&\le \sqrt{2+2\tau}.
\end{aligned}$$

Thus we obtain that $d(\varpi_n\nu_0,\nu_0) = \mathcal{O}\left(n^{-\frac{\beta}{\beta+d+1}}\right)$. □

Proof of Lemmas 4 and 5. Omitted as the proof is similar to Wu et al. (2023). □

Proof of Lemma 6. By Lemma 5, first we have that $N_{[]}(\epsilon,\mathcal{G}_n,\|\cdot\|_\infty) \le \tilde{N}_{[]}(\epsilon,\mathcal{G}_n,\|\cdot\|_\infty)$. By Lemma 2, there exists a constant $c_1 > 0$ such that for arbitrary $\hat{\nu}_1,\hat{\nu}_2 \in \mathcal{V}_n$, we have that

$$\|l(T,\delta,\boldsymbol{x};\hat{\nu}_1) - l(T,\delta,\boldsymbol{x};\hat{\nu}_2)\|_\infty \le c_1\|\hat{\nu}_1 - \hat{\nu}_2\|_\infty,$$

which indicates that as long as $\|\hat{\nu}_1 - \hat{\nu}_2\|_\infty \le \frac{\epsilon}{2c_1}$, we have that $\|l(T,\delta,\boldsymbol{x};\hat{\nu}_1) - l(T,\delta,\boldsymbol{x};\hat{\nu}_2)\|_\infty \le \epsilon$. Thus, we have

$$\tilde{N}_{[]}(\epsilon,\mathcal{G}_n,\|\cdot\|_\infty) \le \tilde{N}_{[]}\left(\frac{\epsilon}{2c_1},\mathcal{V}_n,\|\cdot\|_\infty\right).$$

□

Appendix B. Variational updates of mixture weights

We present more details on updating the variational mixture weights. We have the following closed-form solution of γ that minimizes the KL divergence term

$$\gamma_{1,k} = 1 + \sum_{b=1}^{B}\phi_{b,k}, \quad \gamma_{2,k} = \eta_1 + \sum_{b=1}^{B}\sum_{r=k+1}^{T}\phi_{b,r}, \tag{B.1}$$

for $b \in \{1,\dots,B\}$, where B is the sample size and T is the maximum number of clusters. We then compute the log of posterior probability (i.e., the weighted $\log\phi$) as follows,

$$\begin{aligned}
\log\phi_{i,k} &= \text{const} + \mathbb{E}_{\beta\sim q}[\log\pi_{i,k}] + \mathbb{E}_{\alpha\sim q}[\log\nu_{i,k}]\\
&+ \mathbb{E}[\log p(\mathbf{x}_i)] + \mathbb{H}[q_{\psi_k}(\cdot|z_i=(k),\mathbf{x}_i)],
\end{aligned} \tag{B.2}$$

where π and ν are respectively the assigned weights to the primitive and the heavy-tail distributions.

B.1. Primitive distributions

We provide the density and survival functions of the primitive distributions.

Weibull Distribution.

The density of the Weibull distribution is given by

$$f(x;\lambda,k) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1}e^{-(x/\lambda)^k}.$$

The survival function of Weibull distribution is given by

$$F(x;\lambda,k) = 1 - e^{-(x/\lambda)^k},$$

where $k > 0$ is the shape parameter and $\lambda > 0$ is the scale parameter.

Log-normal Distribution.

The density of the Log-normal distribution is given by

$$f(x|\mu,\sigma) = \frac{1}{x\sigma\sqrt{2\pi}}e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}.$$

The survival function of Log-normal distribution is given by

$$F(x|\mu,\sigma) = \frac{1}{2} + \frac{1}{2}\text{erf}\left(\frac{\ln x - \mu}{\sqrt{2}\sigma}\right),$$

where μ is the shape parameter and $\sigma > 0$ is the scale parameter.

The error function erf(x) is defined as

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt.$$

B.2. Heavy-tail distributions

Log-Cauchy Distribution. We adopt the log-Cauchy distribution as the heavy tail distribution.

The multivariate Gaussian Distribution is defined as

$$p(\boldsymbol{x};\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{n}{2}}|\boldsymbol{\Sigma}|^{\frac{1}{2}}}\exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\}.$$

B.3. KL divergences of two multivariate normal distribution

The KL divergences of two multivariate normal distributions $\mathcal{N}(\boldsymbol{\mu}_1,\Sigma_1)$ and $\mathcal{N}(\boldsymbol{\mu}_2,\Sigma_2)$

$$\begin{aligned}
\text{KL}(\mathcal{N}(\boldsymbol{\mu}_1,\boldsymbol{\Sigma}_1)\|\mathcal{N}(\boldsymbol{\mu}_2,\boldsymbol{\Sigma}_2)) = \frac{1}{2}\Big[&\log\frac{|\boldsymbol{\Sigma}_2|}{|\boldsymbol{\Sigma}_1|} - p + \text{tr}\{\boldsymbol{\Sigma}_2^{-1}\boldsymbol{\Sigma}_1\}\\
&+ (\boldsymbol{\mu}_2-\boldsymbol{\mu}_1)^T\boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_2-\boldsymbol{\mu}_1)\Big].
\end{aligned}$$

Appendix C. More on experiment details

C.1. Dataset details

SUPPORT (Knaus et al., 1995): The SUPPORT (Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments) dataset was derived from a study aimed at constructing a model to predict 180-day survival among critically ill hospitalized patients. It comprises records from 9105 patients, of whom 68.1% died during the follow-up period, with a median survival of 58 days. The study assessed 30 variables, including demographic attributes (such as age, sex, race, education, and income), physiological indicators, and comorbidities.

Missing clinical measurements were imputed using standard normal values, while other missing data were completed using mean imputation for numerical variables and mode imputation for categorical ones.

SYNTHETIC (Nagpal et al., 2022): The SYNTHETIC dataset is artificially generated using the auton-survival[1] GitHub repository. It consists of 5000 samples with 9 features and maintains a censoring rate of approximately 20%. The dataset includes two distinct risk types, enabling analysis of competing risks. Details of the data generation process are provided in later subsection.

MIMIC-III (Johnson et al., 2016): We assess our model using the MIMIC-III v1.4 database[2], a detailed and publicly available clinical dataset. As outlined in Harutyunyan et al. (2019), we extract time-series data for 17 clinical variables from the first 48 hours of each ICU admission. Due to significant missing data, we fill in the gaps using the most recent available value, or a predefined "normal" value if no previous measurement is available (Harutyunyan et al., 2019). Recognizing that missingness patterns may be informative, we add a missingness mask for each clinical variable at each timestamp, resulting in a feature dimension of 34. Censored data and survival times are extracted from the corresponding MIMIC-III tables. After preprocessing, we obtain a total of 17,814 samples.

MIMIC-IV (Johnson et al., 2023): To further evaluate our model, we test its performance on MIMIC-IV v2.2[3], another extensive open-source clinical database. As MIMIC-IV is an updated version of MIMIC-III, we apply a similar preprocessing approach as in Hayat et al. (2022), extracting time-series data for 15 clinical variables from the first 48 hours of each ICU admission. The same imputation method is used, and by adding missingness masks to the original features, we achieve a feature dimension of 30 per timestamp. Notably, MIMIC-IV includes two fewer variables than MIMIC-III, as these variables are absent in all samples. The final preprocessed dataset contains 22,913 samples for survival prediction.

C.2. Hyperparameters

We conduct experiments to select the optimal hyperparameters, including the network size for the representation learning function and the concentration rate for the Dirichlet Process (DP).

As shown in Table C.1, we use the C^{td} as the evaluation metric, run our experiments on the SUPPORT and MIMIC-IV datasets and record the results for six different values of η (we let $\eta = \eta_1 = \eta_2$) under four different test horizons. The results indicate that the concentration rate generally makes no significant impact on the performance of DDPSurv. Among these concentration rates, the default value 10 has a slight advantage over others.

As shown in Table C.2, we use C^{td} as the evaluation metric and conduct experiments on the SUPPORT and MIMIC-IV datasets, recording results for five different network sizes across four test horizons. For the SUPPORT dataset, a network size of 500 performs best, while for MIMIC-IV, a network size of 2000 is optimal. This suggests that the ideal network size varies with the dataset, with moderate sizes generally yielding better performance.

C.3. Generative process for the synthetic dataset

1. Features x_1, x_2 and the base survival phenotypes Z are sampled from `scikit-learn's make_blobs(…)` function which generates isotropic Gaussian blobs:

$[x_1, x_2], Z \sim \texttt{sklearn.datasets.make_blobs(K=3)}$

[1] https://github.com/autonlab/auton-survival.
[2] https://physionet.org/content/mimiciii/1.4/.
[3] https://physionet.org/content/mimiciv/2.2/.

Table C.1
The results (C^{td}) for different model hyperparameter η under SUPPORT and MIMIC-IV datasets. 25th, 50th, 75th and 90th, represent the time horizon of time dependent C^{td}. Best results are in bold.

η	SUPPORT					MIMIC-IV				
	25th	50th	75th	90th	Mean	25th	50th	75th	90th	Mean
0.01	72.75	68.10	66.03	65.81	68.17	77.59	72.57	68.02	64.83	70.75
0.1	72.79	68.13	66.04	65.81	68.19	77.49	72.63	67.2	63.83	70.29
1	74.37	69.00	65.84	65.61	68.71	77.82	73.33	68.10	64.92	71.04
10	74.32	69.01	65.99	65.65	68.74	77.79	73.32	68.16	64.95	71.06
100	73.34	67.98	65.00	64.67	67.75	77.43	72.00	67.16	63.64	70.06
1000	73.35	67.96	65.00	64.65	67.74	77.45	72.31	67.61	64.26	70.41

Table C.2
The results (C^{td}) for different model hyperparameter net size under SUPPORT and MIMIC-IV datasets. 25th, 50th, 75th and 90th, represent the time horizon of time dependent C^{td}. Best results are in bold.

Net Size	SUPPORT					MIMIC-IV				
	25th	50th	75th	90th	Mean	25th	50th	75th	90th	Mean
200	73.95	68.83	65.2	64.9	68.22	77.78	72.48	67.20	63.69	70.29
500	74.32	69.01	65.99	65.65	68.74	77.79	73.32	68.16	64.95	71.06
1000	72.13	67.73	65.84	65.36	67.77	78.08	72.48	67.29	63.87	70.43
1500	70.89	67.17	65.4	65.25	67.18	76.24	73.41	68.13	65.56	70.84
2000	69.23	67.07	65.74	65.22	66.82	78.77	73.23	68.62	65.55	71.54

2. Features x_3 and x_4 are sampled uniformly, whereas the underlying treatment effect phenotypes ϕ are defined according to an L_1-ball:

$[x_3, x_4] \sim \texttt{Uniform}(-2, 2)$

$\phi \triangleq \mathbb{I}\{|x_3| + |x_4| > 2\}$

3. We then sample treat assignments from a Bernoulli distribution:

$A \sim \texttt{Bernoulli}(\frac{1}{2})$

4. Next, the time-to-event T conditioned on the confounders x, latent Z and latent effect group ϕ are generated from a Gompertz distribution:

$T^*|(Z = k, \phi = m, A = a) \sim \texttt{Gompertz}\left(\beta_k^\top x + (-a^m)\right)$

5. Finally, the observed time T is obtained after censoring some of the events and censoring time is chosen uniformly at random upto T^*:

$\delta \sim \texttt{Bernoulli}(\frac{3}{4}), \quad C \sim \texttt{Uniform}(0, T^*)$

$$T = \begin{cases} T^*, & \text{if } \delta = 1 \\ C, & \text{if } \delta = 0 \end{cases}$$

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